

Intake of fiber and fiber-rich plant foods is associated with a lower risk of renal cell carcinoma in a large US cohort



Background: Plant-based and fiber-rich diets high in vegetables, fruit, and whole grains are recommended to prevent cancer and chronic conditions associated with renal cell carcinoma (RCC), such as obesity, hypertension, and diabetes. Diet may play a role in the etiology of RCC directly and/or indirectly. Objective: In a large prospective cohort of US men and women, we comprehensively investigated dietary intake and food sources of fiber in relation to RCC risk. Design: Participants of the NIH-AARP Diet and Health Study (n = 491,841) completed a self-administered questionnaire of demographics, diet, lifestyle, and medical history. Over 9 (mean) years of follow-up we identified 1816 incident cases of RCC. HRs and 95% CIs were estimated within quintiles by using multivariable Cox proportional hazards regression. Results: Total dietary fiber intake was associated with a significant 15–20% lower risk of RCC in the 2 highest quintiles compared with the lowest (P-trend = 0.005). Intakes of legumes, whole grains, and cruciferous vegetables were also associated with a 16–18% reduced risk of RCC. Conversely, refined grain intake was positively associated with RCC risk in a comparison of quintile 5 with quintile 1 (HR: 1.19; 95% CI: 1.02, 1.39; P-trend = 0.04). The inverse association between fiber intake and RCC was consistent among participants who never smoked, had a body mass index [BMI (in kg/m²)] <30, and did not report a history of diabetes or hypertension. Conclusions: Intake of fiber and fiber-rich plant foods was associated with a significantly lower risk of RCC in this large US cohort. This trial was registered at clinicaltrials.gov as NCT00340015.